



OPEN Cognitive bias in generative AI influences religious education

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This study explores the transformative role of generative artificial intelligence (AI) in shaping religious cognition, with particular emphasis on its implications for religious education. By examining the biases inherent in AI-generated content, this research highlights how these biases influence user perceptions and interactions with diverse religious teachings. Through experimental frameworks and pre/post-interaction evaluations, the study reveals that generative AI not only reflects but amplifies cognitive biases, affecting users' understanding of religious doctrines and cultural diversity. The findings underscore the potential of generative AI to act as a double-edged sword in religious education: enhancing personalized learning and cross-cultural understanding while risking the reinforcement of prejudice. These insights call for ethical guidelines and oversight mechanisms in deploying generative AI within religious contexts. This research contributes to the growing discourse on AI ethics and its pivotal role in shaping inclusive and unbiased religious education in the digital era.

Keywords Generative AI, Religious education, Cognitive bias, AI ethics, Experimental design

Generative AI is rapidly emerging as a transformative tool in various domains, including religious education, by leveraging advancements in deep learning and natural language processing technologies. This innovative technology enables the creation of diverse content such as text, images, and audio by analyzing vast datasets^{1,2}. While its applications in personalized content delivery are widely recognized, its implications for religious education have yet to be comprehensively explored. As religions represent both a foundational aspect of cultural identity and a complex system of belief, the integration of generative AI offers both opportunities and challenges for educators and practitioners. By tailoring content to individual learners, AI can enhance engagement and foster deeper understanding of religious doctrines and values.

Traditionally, religious education has relied on structured learning environments mediated by human educators, canonical texts, and experiential practices. However, the advent of the internet and digital technologies has reshaped how religious knowledge is disseminated and consumed³. Digital platforms enable access to diverse religious teachings across geographical and cultural boundaries, promoting inclusivity and cross-cultural understanding. Yet, with the introduction of generative AI, this dynamic has reached new levels of complexity. In religious education, by generating personalized religious content that adapts to individual learners' interests and feedback, AI helps foster a systematic understanding of religious beliefs and practices⁴. Additionally, AI can clarify ambiguities in religious texts, enhancing their interpretability⁵, and further elucidate their cognitive, emotional, and social impacts⁶. AI also encourages learners to engage in dialogue with respect and sensitivity, inspiring religious thought through the introduction of new metaphors and narratives⁷.

Nevertheless, these opportunities may also come with potential challenges. The reliance of generative AI on training data introduces the risk of cognitive bias, which can affect the accuracy and impartiality of religious content. "Cognitive bias" refers to the systematic deviation from objective facts in an individual's judgment due to inherent cognitive patterns or external information influences, leading to irrational or skewed cognitive outcomes. Algorithmic bias, in a certain sense, contributes to this cognitive bias⁸. It includes selection bias, confirmation bias, and measurement bias⁹. For example, confirmation bias causes individuals to seek and interpret information that confirms their pre-existing beliefs, which may reinforce religious convictions¹⁰. On the surface, AI bias may seem to result from faulty algorithms, possibly influenced by data collection and filtering methods; however, a deeper cause lies in the embedded biases of the development team¹¹. If biases embedded in training datasets remain unaddressed, they may lead to skewed representations of religious doctrines, reinforcing stereotypes and hindering interfaith understanding¹². Whether intentional or not, biases in artificial intelligence directly impact the exercise of digital religion or the right to freedom of belief¹³. Affective AI could significantly influence values such as privacy and autonomy¹⁴, with the ultimate goal of reshaping human behavior¹⁵. Such concerns are particularly pertinent in religious education, where the aim is often to

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foster tolerance and mutual respect among diverse beliefs. This dual-edged nature of generative AI underscores the urgency of examining its application within religious contexts, particularly its ability to shape cognition, attitudes, and societal perceptions toward religion.

This study aims to investigate the role of generative AI in religious education, focusing on three key research questions:

Bias in AI-Generated Religious Content: How do training data and algorithms shape the biases inherent in AI-generated religious content, and what are the implications for fairness and objectivity?

Changes in Users' Religious Cognition and Attitudes: How does engagement with AI-generated content influence users' understanding of religious doctrines, and what mechanisms drive these cognitive and attitudinal changes?

Impact on Religious Tolerance and Understanding: Can AI-generated religious content foster interfaith dialogue and tolerance, or does it exacerbate misunderstandings and social conflicts?

By addressing these questions, this research contributes to the growing discourse on the ethical and educational applications of generative AI, emphasizing its transformative potential in religious education while cautioning against its risks. The findings offer a theoretical framework for integrating AI responsibly within educational and religious domains, ensuring inclusivity, accuracy, and fairness.

Literature review

Development and application of generative artificial intelligence

Generative AI refers to a technology that creates new content by learning from and analyzing vast datasets through machine learning algorithms. Unlike traditional artificial intelligence, generative AI can not only analyze existing data but also generate new content akin to human creation. The advancement of this technology has been driven by progress in sophisticated algorithms, such as deep learning, natural language processing, and generative adversarial networks (GANs)¹⁶. These advances enable AI to generate various forms of content, including text, images, and audio. The emergence of generative AI has expanded the application potential of artificial intelligence, particularly in fields such as education, healthcare, and media.

In the field of education, generative AI is progressively becoming a powerful tool for personalized learning. By analyzing data on students' learning behaviors, generative AI can automatically create tailored learning materials that align with each student's proficiency level and needs. Additionally, generative AI is beneficial in language learning, where it can generate dialogue exercises and provide real-time feedback to help students improve their language skills more effectively¹⁷. In religious education, AI can enhance students' understanding and appreciation of different religions by producing diverse instructional materials¹⁸. However, this application necessitates careful oversight by educators to ensure the generated content is accurate and unbiased¹⁹.

In the medical field, generative AI is primarily applied in disease diagnosis, drug research and development, and personalized treatment. This technology enhances the accuracy and efficiency of medical diagnoses while advancing personalized medicine, allowing patients to receive more precise and effective treatments²⁰. Furthermore, generative AI aids in drug discovery by simulating interactions between drug molecules and biological targets, thereby generating potential drug compounds. This approach significantly reduces both the time and cost associated with drug development.

In the media industry, generative AI is utilized for generating news reports, writing articles, and producing video content. News organizations can leverage generative AI to create personalized news content tailored to the needs of diverse audiences. By analyzing users' reading habits and interests, AI can automatically generate content recommendations that align with users' preferences, thereby enhancing engagement and satisfaction. This personalized approach to content generation and distribution improves the efficiency and accuracy of information dissemination. However, it also raises ethical concerns regarding the authenticity of information and the potential for content manipulation.

Influence of generative AI on user cognition

Through frequent interactions with users, generative artificial intelligence (AI) can effectively influence users' cognitive frameworks and information processing methods²¹. During these interactions, generative AI continuously adjusts the content it produces based on user feedback, thereby establishing a dynamic two-way interaction mechanism. This mechanism enables AI to generate personalized and customized content aligned with users' needs and interests, enhancing their sense of engagement and identification. As users interact with generative AI, they gradually accept and internalize the AI-generated information, which in turn shapes their cognitive frameworks²². For example, when users frequently encounter AI-generated news or information on a particular topic, they may develop fixed perceptions or attitudes toward that topic, potentially viewing these AI-generated contents as authoritative sources²³.

By generating and delivering highly customized content, generative AI can gradually influence users' cognitive preferences and attitudinal choices. Through the analysis of users' behavioral data and historical preferences, AI can produce content that closely aligns with users' interests²⁴. Continuous user feedback enables AI to optimize its content generation logic over time²⁵. This interactive mechanism allows generative AI to better understand user needs and deliver content that closely matches their preferences²⁶. This mechanism is also applicable in the context of religious information dissemination. By incorporating user feedback, AI can refine its content generation processes to more effectively convey religious teachings and beliefs²⁷.

However, while personalized content generation enhances user experience, it also risks creating "information cocoons," where users are only exposed to information that reinforces their existing beliefs, thereby neglecting diversity and heterogeneity. This phenomenon can exacerbate cognitive biases and attitudinal polarization, making it difficult for users to encounter alternative viewpoints. Consequently, this limits the breadth and depth of users' cognition²².

The content generation capabilities of generative AI also affect the accuracy and authenticity of users' cognition. Since AI generates content based on its training data, which may include biased, inaccurate, or false information, the generated output can reflect these flaws. When users encounter such biased or misleading content, it may distort their understanding of reality. Furthermore, if AI-generated content contains emotionally charged language, it can influence users' emotional responses to specific topics, subtly altering their attitudes and behaviors²⁴.

Additionally, the self-learning and optimization capabilities of generative AI may amplify its influence on users' cognition. By continuously collecting user feedback, generative AI refines its content generation strategies, enabling it to increasingly align with users' cognitive needs. This iterative optimization process reinforces users' existing cognitive frameworks and attitudes, making AI's influence more persistent and profound. Consequently, the ongoing refinement of AI-generated content can have long-term and far-reaching effects on users' cognition and attitudes²⁸.

Digital prejudice and AI ethics

A notable challenge that generative AI faces in content generation is the presence of digital bias and associated ethical concerns. Since generative AI relies on large-scale data training, this data often reflects the various biases and injustices present in the real world. These biases are particularly problematic when dealing with sensitive subjects such as religion, race, and gender²⁹. When these prejudices are embedded in AI models without identification or correction, generative AI can inherit and amplify these biases in its outputs, negatively impacting users' cognition and presenting serious ethical challenges.

The training data on which generative AI models depend frequently contains social biases from real-world contexts, spanning various aspects such as race, gender, religion, and culture. If this data includes sexist or racist content, the AI may inadvertently reproduce or amplify these biases when generating text or images. This can result in discrimination and unfairness in the generated content. Users may adopt the biases generated by AI without critical reflection³⁰. Such issues are particularly sensitive concerning religious content, where AI-generated outputs may reflect specific religious biases, thereby shaping users' perceptions and attitudes toward different religions. This can exacerbate distrust and conflict between religious groups³¹.

Biases in generative AI are not solely derived from training data; they may also stem from imbalances or biases in model design. Some AI models may favor content that aligns with the values of mainstream culture or dominant groups, thereby marginalizing the perspectives of minority or underrepresented groups. This tendency reinforces existing social biases during information dissemination and can further sideline the voices of vulnerable populations. The persistence of these biases not only undermines fairness in content generation but also poses a threat to social justice and cultural diversity.

In addition to digital bias, generative AI presents numerous ethical challenges. Because generative AI can produce highly realistic content, distinguishing between AI-generated content and human-created content can be difficult for users. This ambiguity heightens the risk of spreading misinformation, particularly in sensitive areas such as religion and politics, which can lead to severe societal consequences. If AI-generated content is misused, it could manipulate public opinion, deepen social divisions, and even incite conflict³².

Furthermore, the opacity of generative AI models complicates accountability. The content generation process often operates as a "black-box," making it difficult for users and regulators to trace the specific reasons behind AI decisions. This lack of transparency obscures the attribution of responsibility when biased or unfair content is generated. Determining who should be held accountable—whether AI developers, data providers, or users—remains a contentious issue. Currently, these questions lack definitive solutions and require further exploration through technological advancements and the development of legal and regulatory frameworks³³.

Application of AI in religious education

AI has opened new pathways for enhancing religious understanding and educational practices, helping to engage younger generations³⁴. Users often turn to generative AI for religious guidance and counseling, driven by personal emotional and spiritual uncertainties^{35,36}, thereby creating interactive, inclusive, and adaptive experiences³⁷, which can transform religious education. In response to this trend, many religious institutions are exploring the integration of digital technologies, such as AI, into religious education, including reforms to Islamic curricula and the introduction of AI-assisted teaching methods³⁸. Generative AI can serve as a tool to support various religious teaching and learning processes³⁹, assisting educators in designing more effective teaching strategies and providing learners with more efficient learning methods³⁷. When used ethically and responsibly, generative AI can also creatively integrate critical thinking into religious education environments⁴⁰ and even reshape religious customs and related behavioral norms⁴¹.

However, in religious education, AI-generated content based on religious narratives or doctrines can exploit cognitive biases, distorting individuals' perceptions of religious truths⁴². This raises concerns about the spread of algorithmic bias, which could exacerbate religious discrimination and inequality⁹. Despite existing research highlighting the risk of bias in generative AI's dissemination of religious content, the dynamics of how its generative mechanisms, influenced by user feedback loops, intensify bias and the differential impact of such bias on religious tolerance remain underexplored⁴³. This study attempts to explore this mechanism through experimental design and aims to identify actionable interventions for promoting inclusivity and sustainability in religious education under the influence of digitalization.

Analytical framework

As shown in Fig. 1, according to the interaction mechanism between generative artificial intelligence and users, we have drawn an analysis framework.

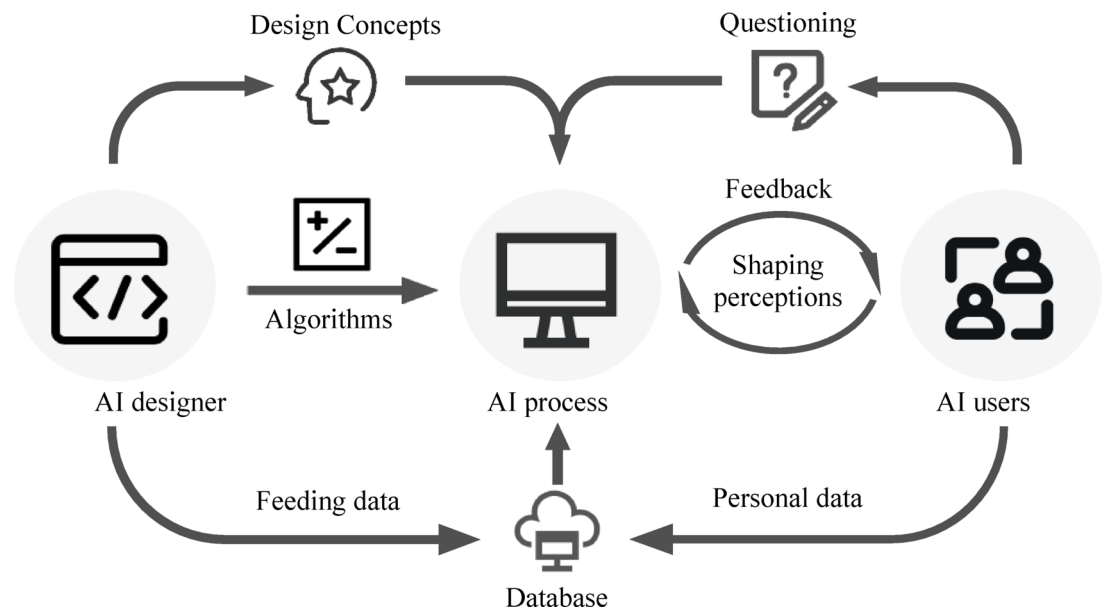


Fig. 1. Analysis framework of the influence of generative AI on user cognition.

Input stage: concept construction, algorithm design and data input

Concept construction involves setting and defining the goals and principles for generating religious content through Generative AI. This process must consider the sensitivity of religious topics, ensuring that the AI-generated content reflects diversity and fairness while avoiding any form of religious bias. For instance, in this study, the AI system was designed with the objective of generating content that remains as neutral as possible, providing objective descriptions of religious themes. However, insufficient attention to ethical considerations and social responsibility during concept construction may lead to religious misunderstandings or conflicts during the dissemination of generated content.

Algorithm design is a core component in the generation of religious content by AI, determining how the system processes and analyzes input data. The models employed in this research include advanced natural language processing frameworks, such as GPT-4, which generate context-specific religious content through large-scale data training. A critical aspect of algorithm design is identifying and mitigating biases in the training data to prevent the amplification of these biases during content generation. Accordingly, adversarial training and bias detection techniques were applied in this study to automatically identify and rectify potential religious biases within the model's output. The primary objective of algorithm design is to ensure that the generated content maintains neutrality in both semantic and emotional expression, avoiding undue favoritism towards any specific religious belief or cultural perspective.

Data input forms the foundation for generative AI content creation, directly impacting the fairness and accuracy of the generated output. The data sources utilized in this study encompass a broad range, including religious texts, social media content, and news reports. These datasets influence how AI interprets and represents religious themes. During the data input phase, rigorous data cleaning and preprocessing are essential to ensure diversity and impartiality. Analysis of the questionnaire data reveals that certain religions may be underrepresented in AI-generated descriptions, reflecting biases inherent in the input data. Therefore, ensuring the fairness and representativeness of input data is paramount to mitigating prejudice during the content generation process.

Processing stage: generation and adjustment

Content generation is the process by which generative AI transforms input data into specific religious information. During this process, AI models generate text or other forms of content based on predefined algorithmic logic. The AI system used in this study, such as models based on GPT-4, can produce seemingly reasonable and coherent religious descriptions derived from extensive training data. These descriptions may include interpretations of religious teachings, retellings of historical events, or portrayals of religious figures. However, because content generation by generative AI relies heavily on training data and algorithmic configurations, the output may reflect hidden biases or tendencies. Research findings indicate notable differences in the portrayal of certain religions within AI-generated descriptions, suggesting that data bias influences the content produced. Therefore, ensuring that AI-generated religious information is fair, accurate, and ethically sound is paramount.

Interaction and feedback is a dynamic process in which generative AI collects user input and adjusts the generated content in real time. In the context of religious content generation, user feedback helps the AI refine its output to better meet user needs. After interacting with AI-generated religious descriptions, users can provide feedback through multiple-choice questions, ratings, or direct commentary. By analyzing this feedback, the AI system can adjust its content generation strategy in real time, ensuring subsequent outputs align more

closely with users' cognitive models and preferences. This feedback-driven adjustment mechanism enables AI to continuously optimize content generation and enhance user experience. However, this mechanism can also lead to AI excessively reinforcing users' existing cognitive biases, exacerbating the information cocoon effect. Research data reveal that users' perceptions of certain religions shifted after interacting with AI-generated content, reflecting the adjustments AI made based on user feedback. Consequently, the design of interaction and feedback mechanisms must balance improving user experience with preserving content diversity and fairness.

Feedback stage: user's cognition change

After users are exposed to religious descriptions generated by AI, they typically express their satisfaction and understanding of the content through scoring, multiple-choice questions, or open-ended feedback. These forms of feedback provide valuable data for evaluating the effectiveness and user acceptance of the generated content. Research findings indicate that in groups where AI-generated content was introduced, users' evaluations of different religions varied, underscoring the pivotal role of user feedback in content generation. By analyzing this feedback, AI can identify issues such as bias or inaccuracies and make appropriate adjustments to improve content fairness and user recognition.

Generative AI dynamically shapes religious cognition through user feedback mechanisms, with its impact manifesting in two areas: cognitive restructuring and attitude polarization. On the cognitive level, AI-generated religious content may expand users' knowledge breadth. However, due to the implicit biases in the algorithm's training data, it is prone to fostering one-sided interpretations, such as misreadings of specific religious terms or the reinforcement of stereotypes. On the attitude level, AI-generated content, guided by emotional biases, may trigger emotional polarization in users' views toward specific religions, leading to a decline in positive evaluations or an increase in neutral attitudes⁴⁴. Additionally, users' trust in AI-generated religious information can influence their attitudes toward traditional religious authority, potentially undermining it. For instance, the study found that in the AI-interference group, users' evaluations of Islam were notably lower compared to those in the non-interference group. This finding suggests that AI-generated content may reinforce biased or misleading perceptions. Such cognitive and attitudinal changes often result from repeated exposure and feedback interactions, potentially exerting a long-term impact on users' understanding.

The bidirectional feedback loop between generative AI and users is fundamentally a dynamic interaction of "data feeding-content generation-feedback reinforcement." AI continuously optimizes the generated content based on users' religious preferences and feedback, while users gradually internalize the implicit values or biases embedded in the AI-generated religious information through repeated exposure. In this loop, AI may inadvertently amplify religious stereotypes present in the training data, while users' positive feedback further solidifies algorithmic biases and cognitive distortions. This ultimately leads to the echo chamber effect, diminishing the diversity and objectivity of religious understanding. In the study, comparisons between the AI-interference group and the non-interference group revealed the significant role of this two-way circulation mechanism in religious content generation. This mechanism not only drives the continuous improvement of AI but is also crucial for bias control.

Methods and data analysis

Research design

In this study, an experimental design was employed to assess respondents' religious cognition through a questionnaire survey. The experiment consisted of two stages: the baseline measurement stage and the intervention stage.

This study employed stratified random sampling, recruiting 1,005 participants through an online platform, covering 12 geographic regions (e.g., East Asia, Southeast Asia, Middle East), with sample sizes allocated according to regional population proportions. The sampling process included geographic stratification (based on World Bank regional classifications) and random selection within each stratum, ensuring balance between the intervention group (502 participants) and the control group (503 participants) across dimensions such as nationality, religious affiliation, gender, age, and educational background. The sample represents diverse religious backgrounds, with primary information sources being online media (38.2%) and social organizations (23.8%), ensuring data uniqueness and ethical compliance.

In the baseline measurement stage, all respondents completed a questionnaire assessing their initial attitudes and cognitive perceptions of various religions before being exposed to generative AI content. This baseline data served as a benchmark for subsequent analysis.

In the intervention stage, respondents were randomly assigned to either the AI-interference group or the non-AI-interference group. In the AI-interference group, participants read religious descriptions generated by generative AI, which contained positive, negative, or neutral views of certain religions. Afterward, they completed the questionnaire again to measure changes in their religious cognition. In contrast, the non-AI-interference group read traditional religious descriptions without AI-generated content and subsequently filled out the same questionnaire.

The core objective of this experimental design was to compare the cognitive and attitudinal changes between the two groups before and after exposure to generative AI content. These comparisons aimed to uncover how generative AI influences users' perceptions of different religions and how potential biases in AI-generated content affect these perceptions.

To ensure the reliability of the results, the study adopted a random assignment method, dividing 1,005 respondents into two groups: 502 participants in the AI-interference group and 503 in the non-AI-interference group. This design maintained the uniqueness of variables between the control and experimental groups, ensuring that the only differentiating factor was exposure to AI-generated content.

The questionnaire evaluated multiple dimensions of religious cognition, including overall impressions of religion, tolerance of doctrine, morality and ethics, societal and lifestyle views, the image of believers, and perceptions of modernity. Responses were recorded using a 5-point Likert scale (ranging from 1 = “strong disapproval” to 5 = “complete approval”), enabling the quantification of respondents’ attitudes toward various religions.

The design and implementation of this study underwent rigorous ethical review. The research plan, detailing the experimental design, participant recruitment, data collection, and analysis methods, was submitted to and approved by the Ethics Committee. Particular emphasis was placed on ethical considerations related to the sensitive nature of religious cognition. The Ethics Committee evaluated potential risks, such as the psychological impact of AI-generated content or the potential for reinforcing religious biases. The research team implemented measures to mitigate these risks. Participants were fully informed about the purpose and procedures of the study, their rights as participants, and the voluntary nature of their involvement. Consent was obtained prior to completing the questionnaire, with assurances that all data would remain confidential and no personal information would be collected or disclosed. All methods employed in this study were conducted in strict accordance with relevant guidelines and regulations, and necessary approvals and authorizations were obtained where required.

Experimental process

In the baseline measurement stage, all respondents completed a questionnaire to assess their initial cognition and attitudes toward various religions. This initial questionnaire provided benchmark data for subsequent analysis. In the intervention stage, respondents in the experimental group completed the questionnaire again after being exposed to religious descriptions generated by generative AI. These AI-generated descriptions encompassed positive, negative, and neutral views, allowing for an evaluation of the impact of AI-generated content on religious cognition. The questionnaire design also included open-ended questions to enable respondents to articulate their personal views on the religious descriptions (See Fig. 2).

Stage 1: Baseline measurement

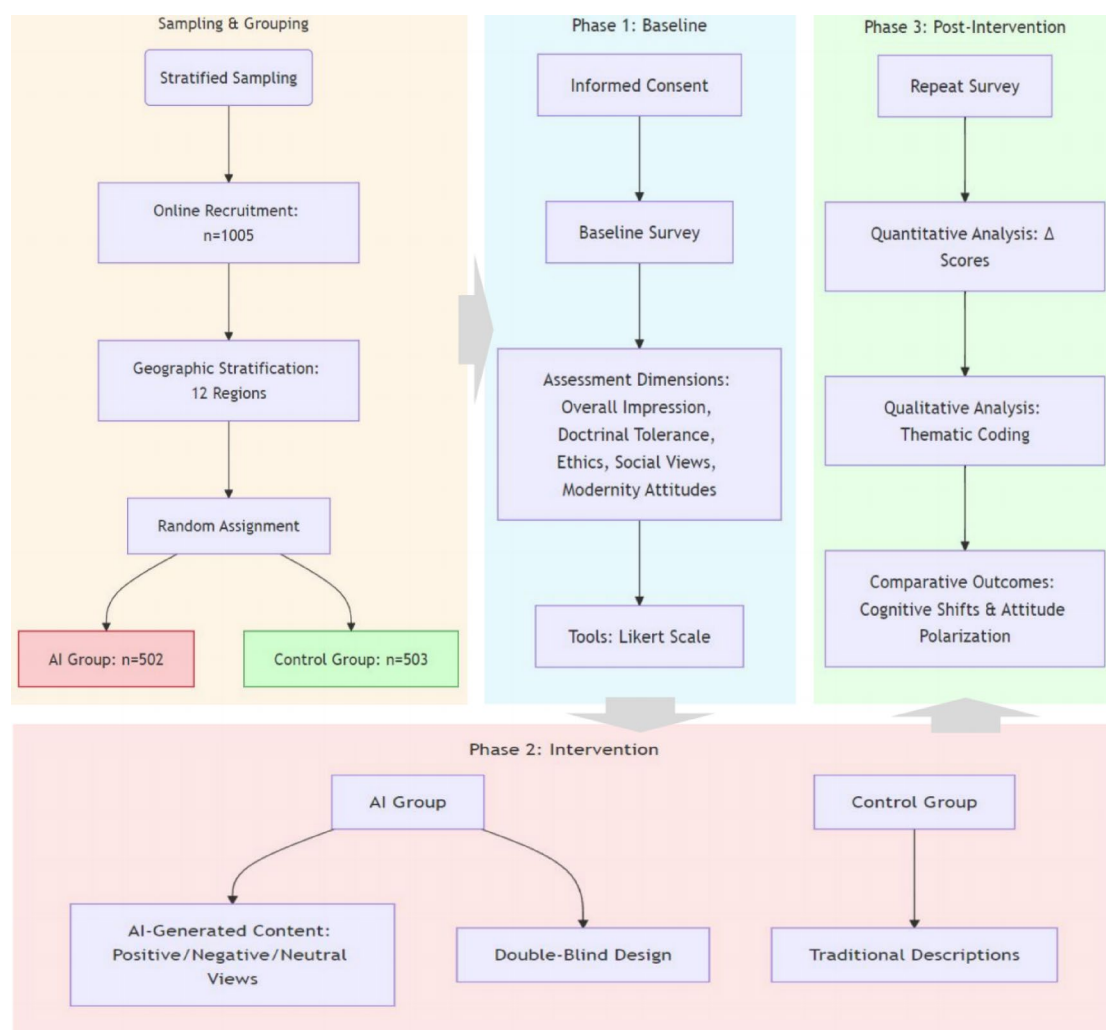


Fig. 2. Experimental design process of this study.

The first stage of the experiment involved a baseline measurement to capture respondents' religious cognition and attitudes before exposure to generative AI-generated content. A total of 1,005 respondents completed a questionnaire assessing various dimensions of religious perception, including overall impressions of different religions, tolerance of doctrines, views on morality and ethics, societal and lifestyle beliefs, perceptions of believers, and attitudes toward modernity. Responses were recorded on a 5-point Likert scale (1 = "strong disapproval" to 5 = "complete approval"). This stage provided a comprehensive benchmark for comparing subsequent cognitive and attitudinal changes.

Stage 2: Intervention implementation

Following the baseline measurement, respondents were randomly assigned to one of two groups: the AI-interference group and the non-AI-interference group. In the AI-interference group, participants read religious descriptions generated by generative AI, which included content with positive, negative, and neutral perspectives on various religions. These descriptions simulated the diverse outputs that AI systems might produce in real-world applications. Respondents in the non-AI-interference group read traditional religious descriptions that did not involve AI-generated content.

To ensure that responses were not influenced by external factors, participants were unaware of the source of the content they read. This design allowed for an unbiased evaluation of the unique impact of generative AI content on religious cognition. It is important to note that, in order to control variables, we simulated only a single cycle of interaction between humans and AI, without conducting repeated dialogues. As a result, the study may not effectively validate the reciprocal shaping and feedback mechanisms between AI and humans.

Stage 3: Intervention measurement

After the intervention, all respondents completed the same questionnaire again to assess any cognitive and attitudinal changes resulting from exposure to generative AI content. This post-intervention survey evaluated the same dimensions as the baseline measurement, including overall impressions of different religions and attitudes toward doctrines, morality, society, and modernity. Additionally, the questionnaire included open-ended questions that allowed respondents to provide detailed feedback and specific reactions to the religious descriptions they encountered.

This multi-stage design facilitated a rigorous comparison of changes between the AI-interference and non-AI-interference groups, revealing the potential influence of generative AI on religious cognition and attitudes.

Generative AI content generation

AI content generation relies on a diverse range of data sources, including religious texts, social media content, and news reports. These data have undergone preprocessing steps such as noise removal, standardization, and de-biasing. The dataset encompasses major world religions, including Christianity, Islam, Buddhism, Hinduism, and Judaism, covering aspects such as religious teachings, historical events, contemporary religious practices, and societal views.

The model employed in this study is based on advanced natural language processing technology, specifically GPT-4. ChatGPT was selected as the experimental tool for this study based on several considerations. The focus of this research is to explore the impact of the interaction mechanisms between generative AI and users on religious cognition, rather than comprehensively evaluating the performance differences of all AI models. ChatGPT (GPT-4) was chosen as the experimental tool because it is currently the most widely used general model with the strongest interactivity for generating religious content, and its underlying architecture represents the mainstream technological approach of generative AI⁴⁵. Furthermore, this model is known for its cultural biases, with training data that includes multilingual religious texts. Several studies have confirmed that its output exhibits systemic cultural skew, such as the 1.7 times higher semantic association between Islam and "violence" compared to other religions⁴⁶. During the content generation process, the AI system produces three types of religious descriptions: positive, negative, and neutral. Positive descriptions highlight the beneficial influence and social contributions of religion, negative descriptions address religious conflicts or adverse events associated with religious practices, and neutral descriptions aim to remain objective, offering factual religious information.

To ensure the accuracy and rationality of the generated content, the religious descriptions produced by the AI were subjected to multiple rounds of review before being used in the experiment. During the implementation phase, respondents in the AI-interference group were presented with the AI-generated content in text form. Each participant randomly read one piece of AI-generated content, which was selected based on their religious background or interests. This approach aimed to assess the impact of different AI-generated descriptions on respondents' religious cognition. The research data indicates that, within the AI-interference group, content with varying tendencies influenced participants' religious cognition, further substantiating the potential role of AI-generated content in shaping users' attitudes.

Data analysis

Data preparation

The collected data from the 1005 questionnaires were thoroughly cleaned and reviewed, focusing on missing values, outliers, and inconsistent responses. For missing values, both deletion and interpolation methods were employed to maintain the integrity of the dataset. Outliers were identified by establishing reasonable value ranges and were either corrected or removed as necessary. All questionnaire responses were converted into numerical codes, with responses on the 5-point Likert scale being coded from 1 to 5 for ease of subsequent statistical analysis.

Additionally, demographic information (e.g., gender, age, nationality, and religious affiliation) was classified and coded to account for potential confounding variables in the analysis. To ensure comparability across variables, all data were standardized. The reliability test results for the research questionnaire are presented in

Items	Alpha	Number
Christianity	0.826	7
Islamism	0.831	7
Buddhism	0.805	7
Hinduism	0.829	7
Judaism	0.726	7
Entirety	0.953	35
KMO Sampling suitability quantity	0.973	
Bartlett sphericity test	Approximate chi-square	9579.538
	freedom	595
	significance	0.000

Table 1. Alpha and KMO test.

Characteristic	Option	Frequency	Percentage
Nationality	North American countries	31	3.1
	Oceania countries	25	2.5
	Southeast Asian countries	168	16.7
	East Asian countries	346	34.4
	Russia and Eastern European countries	43	4.3
	Latin American countries	39	3.9
	Southern African countries	56	5.6
	South Asian countries	109	10.8
	European countries	48	4.8
	other	19	1.9
	Middle East and North Africa countries	64	6.4
	Central Asian countries	57	5.7
Religious belief	Taoism	96	9.6
	Buddhism	239	23.8
	Christianity (Catholic/Protestant/Orthodox)	185	18.4
	Others (other religions/folk religions)	22	2.2
	An infidel/agnostic/atheist	154	15.3
	Islamism	201	20
	Hinduism	69	6.9
	Judaism	39	3.9
Social relationship background (information channels about religion)	Family	65	6.5
	Friend	170	16.9
	Social organizations or institutions	239	23.8
	Community or village	102	10.1
	Books	45	4.5
	Network media	384	38.2

Table 2. Basic information of respondents.

Table 1, with the Cronbach's alpha for all religious categories exceeding 0.7. This indicates that the questionnaire demonstrates good reliability and validity.

For validity evaluation, KMO value measurement method is adopted. The closer the KMO value is to 1, the higher the validity of the questionnaire is. As shown in Table 2, the overall KMO value of the questionnaire is 0.973, and the chi-square value of the bartlett sphericity test is 9579.538, which is lower than 0.05, indicating that the questionnaire has good validity.

Descriptive statistical analysis

A total of 1005 valid questionnaires were collected, with respondents representing diverse geographical regions and cultural backgrounds, including East Asia, Southeast Asia, North America, Oceania, the Middle East, and North Africa. The sample varied in terms of gender, age, education, and religious affiliation. During the recruitment process, questionnaires were distributed through online survey platforms and social media to ensure broad representation across different demographic groups. The majority of respondents were aged between 25 and 34 years, with the predominant education level being undergraduate or junior college (See Table 2; Fig. 3).

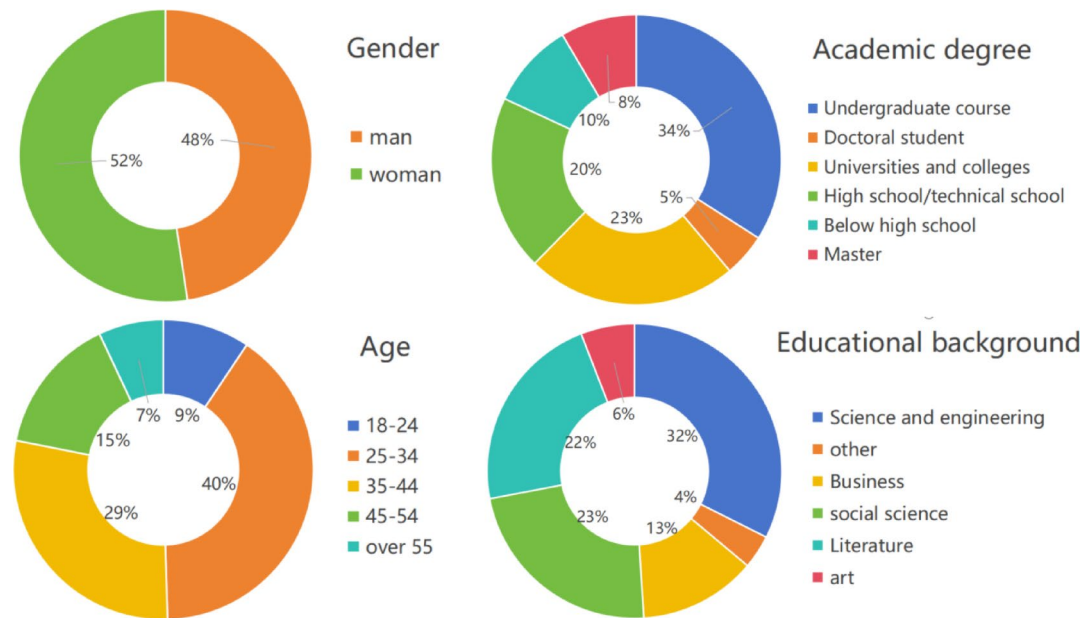


Fig. 3. Background information of respondents.

	Groups	Case number	Average	Std. deviation	t	p
Christianity	Non-Intervened group	503	3.40	0.77	- 7.197	0.000
	AI-Intervened group	502	3.71	0.60		
Islam	Non-Intervened group	503	3.22	0.63	7.718	0.000
	AI-Intervened group	502	2.99	0.27		
Buddhism	Non-Intervened group	503	3.32	0.73	- 0.824	0.410
	AI-Intervened group	502	3.35	0.47		
Hinduism	Non-Intervened group	503	3.19	0.63	2.542	0.011
	AI-Intervened group	502	3.11	0.37		
Judaism	Non-Intervened group	503	3.26	0.58	- 4.101	0.000
	AI-Intervened group	502	3.40	0.52		

Table 3. Respondents’ evaluation of different religions.

The religious backgrounds of the participants include Buddhism, Christianity, Islam, other religions, as well as individuals with no religious affiliation.

In this study, the collected questionnaire data were analyzed using descriptive statistics. Basic statistics, including the mean, standard deviation, minimum value, and maximum value of each index, were calculated to provide a preliminary understanding of the distribution of respondents’ religious cognition and attitudes. Based on a 5-point Likert scale, the average religious scores across different dimensions (such as overall impression, religious tolerance, morality and ethics, etc.) were computed. For clarity, we combined the seven cognitive measures related to each religion (including religious history, doctrinal tolerance, religious ethics, social influence, impression of followers, religious modernity, and overall impression) from the questionnaire. The average score for each religion’s cognitive status was then calculated and used for comparative analysis, as shown in Table 3.

We conducted a correlation analysis to examine the relationship between respondents’ religious cognition and various background factors, including nationality, gender, educational background, religion, and social context. However, after verification, no significant correlation was found between these background factors and religious cognition. To further assess the impact of generative artificial intelligence on different groups, an independent samples t-test was employed. This method compares religious cognition between the group exposed to AI-generated content and the group without AI interference. The analysis determines whether generative AI content leads to cognitive differences between the two groups.

To enhance the persuasiveness of this study regarding potential biases in AI-generated texts, we compared texts from the experimental group (AI-generated) and the control group (human-generated) through content analysis. A set of human-written religious texts was selected to match the AI-generated texts in theme, length, and information richness. We then applied Thematic Analysis and Sentiment Analysis, focusing on Sentiment Polarity, Stereotype & Association, and Information Diversity.

Using the Bias Benchmarking Framework, we identified biases in categories such as Cultural Bias, Sentiment Bias, and Omission Bias through a combination of human annotation and computer-assisted analysis. The results showed that AI-generated texts exhibited more significant biases in sentiment and stereotypes compared to human-written texts. For example, AI-generated texts on Islam contained 1.5 times more references to “conflict,” while Christian texts featured more positive terms like “love” and “forgiveness.” These findings support the hypothesis that generative AI in religious content creation can induce cognitive biases, influencing users’ attitudes.

For Christianity, the group without AI interference had an average evaluation score of 3.4, while the AI-interfered group scored 3.71. The presence of AI interference resulted in a increase of 0.31 points in the evaluation of Christianity ($t = -7.197, p < 0.001$). In contrast, for Islam, the AI-interfered group’s evaluation dropped by 0.23 points ($t = 7.718, p < 0.001$). The evaluation of Buddhism remained largely unchanged ($t = -0.824, p > 0.05$), while Hinduism saw a slight decrease of 0.07 points, with a difference observed in the evaluation of Hinduism due to AI interference ($t = 2.542, p < 0.05$). For Judaism, the AI-interfered group’s evaluation increased by 0.14 points, different from the group without AI interference ($t = -4.101, p < 0.001$). These results suggest that AI-generated content influences respondents’ evaluations of certain religions. Specifically, while the evaluation of Christianity increased, the evaluation of Islam decreased, highlighting the potential biased influence of generative AI in the dissemination of religious information.

To explore whether generative AI content influences religious cognition through intermediary variables such as emotional response and perceived credibility, we conducted a structural equation modeling (SEM) analysis. The results indicate that generative AI content indirectly impacts respondents’ religious attitudes and cognition by enhancing emotional responses and perceived credibility.

Generative AI has a notable impact on users’ religious cognition across six dimensions: religious tolerance, morality and ethics, society and life, the image of Christianity, and modernity. Table 4 presents the differences between the experimental and control groups across these six dimensions. Both the non-interfered and AI-interfered groups showed a positive evaluation of Christianity overall, while Islam received lower overall ratings. Buddhism scored higher in the dimensions of morality and ethics, while Hinduism received higher scores in religious history and doctrinal tolerance.

Figure 4 provides a more intuitive representation of the impact of AI on users’ religious cognition. After reading AI-generated descriptions of different religions, users’ cognition and evaluations of these religions shifted. Specifically, the overall perception of Christianity and Judaism became more positive, while the perception of Islam became more negative. The subjects’ understanding of Hinduism declined slightly, whereas their perception of Buddhism remained relatively balanced. This shift can be attributed to the design concepts embedded in AI systems and the inherent biases within the databases used for training. Currently, mainstream generative AI models are influenced by internet data predominantly from the United States, which may explain the higher recognition of Christianity. The prevalence of religiously biased content in U.S. online media, particularly an implicit unfavorable attitude towards Islam, likely contributes to this bias, which may affect the training of artificial intelligence systems.

Data analysis indicates that, in the baseline measurement, respondents’ evaluations of different religions were relatively neutral. However, after the intervention, the AI interference group exhibited noticeable shifts in religious attitudes. Paired sample t-tests revealed that exposure to generative AI content was associated with a slight increase in the respondents’ evaluation of Christianity, while their evaluation of Islam decreased. This suggests that generative AI content has a biased impact on religious cognition. The results of the independent sample t-test further confirmed the significance of this effect. Specifically, in the AI interference group, respondents’ religious attitudes differed from those in the non-interference group. After controlling for confounding variables, multiple regression analysis demonstrated that AI-generated content independently influences religious cognition. This effect is particularly pronounced through mediators such as emotional response and information credibility,

Evaluation dimensions	Groups	Christianity	Islam	Buddhism	Hinduism	Judaism
Religious history	Non-Intervened group	3.384	3.215	3.245	3.33	3.29
	AI-Intervened group	3.691	2.982	3.44	3.108	3.412
Doctrinal tolerance	Non-Intervened group	3.368	3.256	3.33	3.306	3.274
	AI-Intervened group	3.695	3.012	3.313	3.122	3.402
Morality and ethics	Non-Intervened group	3.435	3.185	3.378	3.149	3.199
	AI-Intervened group	3.735	2.984	3.371	3.112	3.428
Society and life	Non-Intervened group	3.366	3.27	3.326	3.165	3.278
	AI-Intervened group	3.725	2.986	3.271	3.137	3.367
Image of the faithful	Non-Intervened group	3.421	3.227	3.286	3.087	3.266
	AI-Intervened group	3.723	2.962	3.305	3.086	3.404
Modernity	Non-Intervened group	3.384	3.205	3.332	3.173	3.229
	AI-Intervened group	3.773	2.984	3.363	3.092	3.369
Overall impression	Non-Intervened group	3.429	3.213	3.316	3.115	3.29
	AI-Intervened group	3.655	3.004	3.375	3.084	3.436

Table 4. Respondents’ evaluation of different dimensions under the influence of AI (average).

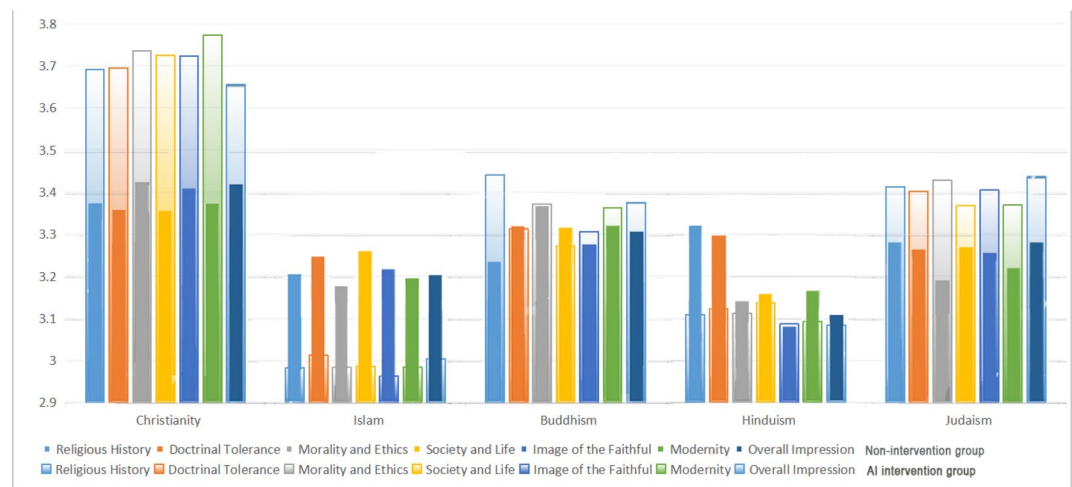


Fig. 4. Influence of AI intervention on respondents' religious cognition.

which amplify the potential biases in AI-generated content. These findings suggest that generative AI may reinforce existing prejudices in the dissemination of religious information, with a profound impact on users' religious cognition.

This study found that AI-generated content exhibits a tendency to influence the perception of different religions, which could have profound implications for religious education and religious inclusivity. In religious education, prolonged exposure to biased AI content may lead to either positive or negative perceptions of certain religions, potentially diminishing the diversity of classroom discussions and hindering fair understanding of religious history and belief systems. For instance, the increased cognitive scores for Christianity and Judaism may enhance their recognition within educational environments, while the decline in the cognitive scores for Islam could reinforce existing societal stereotypes, reducing students' inclusivity toward its culture and faith. As AI becomes a significant tool for information acquisition, this bias may affect interfaith dialogues and, on a broader societal level, shape the cognitive divides between different groups.

Discussion

Generative AI's role in shaping religious education and cognition

Research indicates that generative AI can induce permanent changes in users' religious cognition through repeated exposure² and information reinforcement⁴⁷. When users encounter specific religious content generated by AI multiple times, it may gradually influence their overall impression and attitude toward that religion. This effect is especially pronounced when the generated content contains implicit biases or tendencies, as users may unconsciously internalize this information and perceive it as an objective representation of the religion. This study shows that users in the AI interference group developed a more negative attitude toward certain religions after repeated exposure to AI-generated content, suggesting that AI-generated prejudice may become ingrained in users' long-term memory, leading to a lasting cognitive impact. Data analysis shows that the AI interference group's rating of Christianity increased by 0.31 ($t = -7.197, p < 0.001$), while their rating of Islam decreased by 0.23 ($t = 7.718, p < 0.001$), highlighting the bias in generated content and its impact on religious cognition.

This persistent cognitive change is not only evident at the individual level but may also extend to broader social groups through communication and interaction⁴⁸. When users regard AI-generated religious information as authoritative or accurate, they may propagate these views in social interactions, thereby amplifying personal cognitive shifts to a larger social context. This long-term cognitive shaping, particularly in sensitive areas such as religion, can have profound effects on society and culture.

Algorithmic design and data input in religious content creation

The algorithm design and data input of generative AI directly determine the bias and accuracy of its generated content. The core of the algorithm is to produce content that aligns with specific contexts and needs by learning from large datasets. However, this process is not entirely neutral⁴⁹. Research has shown that if the training data contain biases or tendencies, these biases are internalized by the algorithm and can be amplified during content generation⁵⁰. For example, if the training data emphasize negative events or stereotypes associated with a particular religion, the AI-generated content may disproportionately highlight these negative aspects, thereby shaping users' perceptions of that religion. Data analysis reveals that generative AI tends to reflect biases when processing religious content⁷. For instance, respondents in the AI interference group rated Islam significantly lower than those in the non-interference group, illustrating how algorithmic biases, influenced by biased training data, can be passed on to users⁵¹. The influence of algorithm design and data input on AI-generated content is profound, shaping both the quality of the content and subtly affecting users' religious cognition.

Continuous feedback loops: risks and opportunities

Generative AI not only impacts users' religious cognition during the initial content generation but also through its continuous interaction and feedback mechanism, which constantly adjusts and optimizes the generated content. This dynamic process can be conceptualized as a two-way circulation mechanism of self-learning and adjustment. User feedback plays a crucial role in AI's self-adjustment. After interacting with AI-generated religious content, users provide feedback based on their personal understanding and feelings, and the AI adapts the content accordingly to better align with users' expectations⁵². However, this mechanism may also lead AI to further reinforce users' existing religious prejudices⁵³. For instance, if a user holds a negative attitude toward a religion, AI may generate content that aligns more closely with this bias, thereby creating a self-reinforcing cycle of prejudice.

This research shows that users in the AI interference group may develop more negative attitudes toward certain religions after repeated interactions with generative AI, indicating that the AI further reinforced these negative perceptions through its dynamic feedback mechanism. While this feedback loop enhances the user experience, it may also lead to an "information cocoon" effect, where users are only exposed to information that aligns with their pre-existing beliefs, further deepening their religious prejudices⁵⁴.

Implications for religious education and ethical considerations

These findings may offer valuable insights for religious education. Generative AI could be leveraged to create personalized and inclusive religious educational materials that foster a deeper understanding of diverse beliefs⁵⁵. Misunderstandings of history and doctrine can lead to a superficial understanding of religion⁵⁶. Ensuring that AI accurately and impartially conveys religious history and doctrine, while respecting the uniqueness of each religion, is a crucial consideration for religious educators⁵⁷. By identifying and mitigating biases in algorithm design and training data, educators and developers can ensure that AI-generated content serves as a tool for promoting tolerance and critical thinking rather than reinforcing prejudice.

Moreover, ethical oversight and transparency in algorithmic processes cannot be overlooked. Reliable AI systems should adhere to ethical principles and values⁵¹. Developers must disclose the principles behind AI content generation to allow for external auditing and accountability²⁷. Incorporating diverse datasets and fostering collaborative efforts between technologists and religious scholars can further enhance the inclusivity and fairness of AI-generated content. Additionally, fostering awareness among users about potential biases in AI-generated religious content can empower them to critically engage with the material, mitigating the risks of cognitive bias reinforcement.

Overall, while generative AI presents challenges in managing biases and ethical concerns, its potential to transform religious education and communication is undeniable. By addressing these challenges, we can harness AI's capabilities to foster greater understanding, tolerance, and engagement in religious discourse.

Conclusions

This study demonstrates that generative AI may influence users' religious cognition, with particular emphasis on the biases inherent in algorithm design and data input. The research reveals that when generating religious content, AI may inadvertently amplify hidden biases present in the training data, which in turn shapes users' attitudes and perceptions of specific religions. Additionally, the user feedback mechanism reinforces these biases by aligning the generated content with users' existing cognitive preferences, creating a self-reinforcing cycle. This two-way feedback loop highlights the complexity and potential risks of generative AI in the dissemination of religious information, emphasizing the need for careful identification and control of biases in its technical application.

The findings add to the broader discussion on religious education by highlighting both the potential benefits and challenges of generative AI. On one hand, the technology's ability to tailor educational content provides unprecedented opportunities for fostering interfaith understanding, critical thinking, and inclusivity in religious education. On the other hand, the risks of bias and prejudice necessitate proactive measures, including the development of ethical guidelines and transparent algorithmic processes.

This research highlights the importance of interdisciplinary collaboration among AI developers, educators, and religious scholars to ensure the responsible application of AI technologies in religious education. By fostering awareness of the biases inherent in AI systems and encouraging critical engagement with AI-generated content, educators can empower learners to navigate religious information thoughtfully and inclusively.

Generative AI tools should neither be demonized nor uncritically embraced⁵⁸. AI's strength lies in supplementing, not replacing, educators⁵⁹. Its development must be mindful of religious diversity and ethical standards^{60,61}. AI can foster critical thinking and creativity, enabling educators to engage students in religious studies thoughtfully and inclusively. Additionally, creating a hybrid AI-based learning environment and adopting outcome-based education (OBE) is vital for integrating digital technologies into religious education^{62,63}, promoting a truly harmonious integration of religious education and digital technologies—"Tactful Harmony"⁶⁴.

This study acknowledges several limitations. In terms of research design, the reliance on a single cognitive experiment to measure users' cognitive attitudes, rather than simulating a complex human-AI dialogue environment, may not accurately reflect cognitive biases. The experiment measures short-term cognitive changes, whereas religious attitudes are often shaped by long-term socio-cultural influences. Therefore, future research could employ longitudinal methods to examine the long-term impact of AI-generated content on religious cognition. Additionally, the design of the experiment limits the explanatory power of the "interactive feedback mechanism" discussed in the paper.

Regarding the collection of users' cognitive tendencies, the study used a series of Likert-scale multiple-choice questions, which may not capture the complexity of respondents' thoughts on these variables⁶⁵. Future survey

methods could include in-depth interviews to better explore genuine insights related to religious beliefs and other spiritual matters.

In terms of research tools, the experimental design relied on a single model (GPT-4), and the biases in AI-generated content are constrained by its training data and algorithmic logic. Different models, training data, or prompts may influence the content's biases and thus impact the generalizability of the study's findings. Additionally, users' backgrounds, religious stances, and cultural environments may affect their interpretation of AI-generated content, a factor that was not thoroughly controlled in this study.

Future research could further explore the optimized application of artificial intelligence in religious education, such as how to adjust models to reduce bias and enhance content fairness. Additionally, studies could investigate personalized religious education, analyzing how AI can adapt to learners' needs while maintaining objectivity. Longitudinal research should be employed to examine the long-term impact of AI on religious cognition, avoiding an exclusive focus on short-term attitude changes. Lastly, future studies could explore AI's role in fostering religious inclusivity education, evaluating whether it can promote cross-religious understanding through diverse narratives and interactive learning, thereby genuinely advancing interfaith dialogue and tolerance.

Data availability

Data is provided within the supplementary information files.

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Author contributions

Conceptualization, W.S.; methodology, W.S. and J.Z.; investigation, J.Z., Y.L.; resources, J.Z., Y.L. and W.S.; data curation, J.Z. and W.S.; writing—original draft preparation, J.Z., W.S.; writing—review and editing, J.Z. and W.S.; visualization, J.Z. and W.S.; supervision, W.S. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

This study was reviewed and approved by the School of International Relations Ethics Committee at BISU (Approval No. 2025-001). All participants provided written informed consent prior to participation. The research was conducted in accordance with the ethical standards of the China and the Declaration of Helsinki.

Informed consent

All participants in this study provided informed consent prior to their participation. Participants were fully informed about the purpose of the study, the procedures involved, and their rights as participants. They were assured that their participation was voluntary and that they could withdraw from the study at any time without any negative consequences. Additionally, it was clearly communicated that no personal identifying information would be collected or disclosed, ensuring the confidentiality and anonymity of all responses.

Additional information

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